

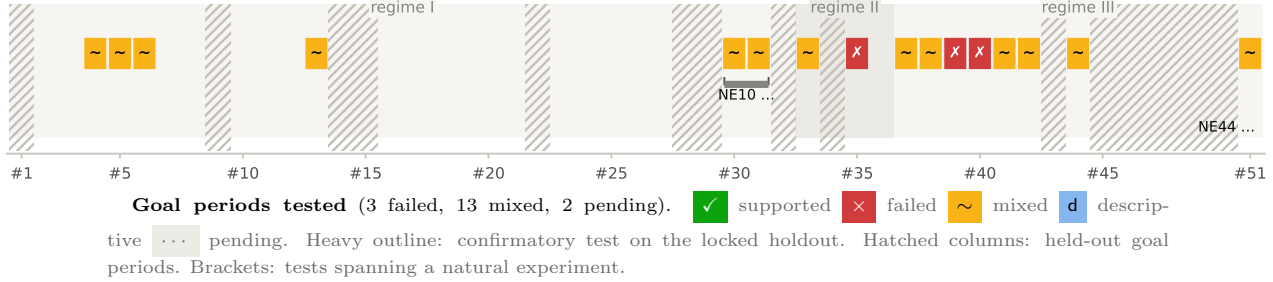
H30: An operator-susceptibility gauge χ_{op} : how steerable is the swarm today?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 42% Faithfulness 2/5 Usefulness 2.5/5

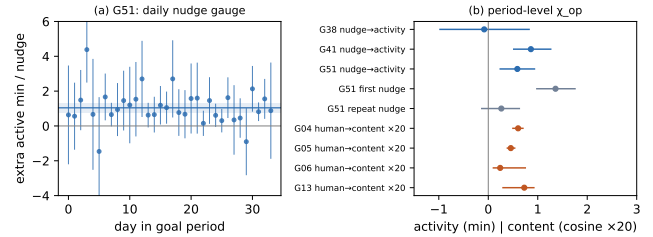
Bottom line. Operator messages have a measurable period-level susceptibility (nudges move the target’s activity, human messages move content), but day-to-day steerability is not a trackable state.

level: hypothesis · pipeline checks 84% · updated 2026-10-03



WHAT HAPPENED

How much does one operator message move the agents who see it, and can that number be tracked day by day? We tested 16 non-holdout goal periods (regimes I–III), after a synthetic validation that found and fixed three estimator biases. *Activity*: in G51 (875 nudges) a nudge adds 0.59 [0.24, 0.93] active minutes in the next 30 min for its target (1.04 with a day fixed effect). First nudges add 1.36; repeats 0.26; room-mates ≈ 0 . G38 is null and the 3–5-day periods are too small. *Content*: human messages turn recipients’ next statements toward their new content: +0.026 cosine for room-mates, +0.046 for named agents, in 6 of 8 periods. Their activity effect is ≈ 0 . *Daily gauge*: mostly noise. Day-to-day variation is weak (G51 $p = 0.08$, $R_1 = 0.28$), does not persist (lag-1 -0.09), and tracks neither goal age, context fill nor lab. H04’s nudge effect (1.66) is inflated by its isolation rule. Holdout not run.



Observables. (a) The G51 daily nudge gauge: extra active minutes per nudge for the target, with day fixed effect, 95% CIs, and the period mean and band. Day-to-day scatter is about the size of the daily CIs: no trend, no persistence. (b) Period-level χ_{op} : nudge→activity in the three largest nudge periods (pre-registered model); first vs repeat nudges in G51 (day FE); human→content in the human-dense periods (cosine $\times 20$).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a set of corrections rather than as a gauge. Someone steering an agent swarm should not watch a daily “steerability” number: at village message rates it is a period constant plus noise, and it does not age. What does carry over: one nudge to an idle agent buys about a minute of extra work, and only the first one does. Repeats and nudges sent into a swarm-wide lull buy almost nothing, and room-mates ignore other agents’ nudges. To change *what* agents write, say something new and name the agent; that moves content, not activity. The methods lesson may matter more: conditioning on whether the nudger fires again silently biased H04’s estimate. The effects are small, single-embedding, and unconfirmed on the holdout.

PHYSICAL MODEL

Models 01 and 11: linear response to impulsive fields. Each operator message k is a field pulse on its recipients i . The activity spin is $n_i(t) \in \{0, 1\}$. Content state: a whitened statement embedding s_i . Message direction u_k , projected off the span of the recipient’s last 8 statements ($u_k^\perp$). χ_{act} : the 30-min integrated activity response, from a local projection with matched pre-history strata α_s , day effects δ_d and kick counts K . χ_{con} : post-message alignment relative to same-kind placebo messages p . $\chi_{\text{op}}(d)$: the day mean of per-kick responses r_k .

$$Y_i(m) \equiv \sum_{\tau=1}^{30} n_i(m + \tau) = \alpha_{s(i,m)} + \delta_d + \sum_c \chi_{\text{act}}^c K_i^c(m) + \beta \cdot K_i^{\text{adj}}(m) + \varepsilon$$

$$\chi_{\text{con},k} = \bar{s}_i^{\text{post}} \cdot u_k^\perp - \langle \bar{s}_i^{\text{post}} \cdot u_p^\perp \rangle_p, \quad u^\perp = \frac{(1 - P_{\text{pre}})u}{\|(1 - P_{\text{pre}})u\|}$$

$$\chi_{\text{op}}(d) = \frac{\sum_{k \in d} w_k r_k}{\sum_{k \in d} w_k}, \quad R_1 = 1 - \frac{\text{Var}_{\text{perm}}[\chi_{\text{op}}(d)]}{\text{Var}[\chi_{\text{op}}(d)]}$$

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RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1	G51 0.59 [0.24, 0.93] (FE 1.04); G38	G51 only
nudge→target	−0.08; 4/9 small > 0	
P2 bystanders ≈ 0	G51 0.13 [−0.11, 0.38]; G38 −0.54	G51 only
P3	≤ 0 in regime I; named never > room	×
human→activity		
P4	0.026 [0.021, 0.030]; named 0.046; 6/8	✓
human→content		
P5	G51 0.003 [0.000, 0.005]	~
nudge→content		
P6 no daily state	G51 $p = 0.08$, $R_1 = 0.28$, lag-1 −0.09	✓
P7 no aging	G51 −0.019 [−0.049, 0.012]/day	✓
P8 context saw-tooth	early 0.94 vs late 1.00 (G51)	×
P10 family	no Holm-significant lab	✓
P11 nulls	G51 pass; G38 placebo −0.59	mixed

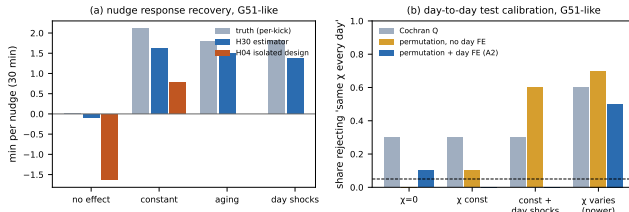
GOAL PERIODS IN DETAIL

G04 ~ mixed — content 0.030*, named 0.033; human activity ≈ 0
G05 ~ mixed — content 0.023*, named 0.046; human activity ≈ 0 ; low power
G06 ~ mixed — content 0.012*, named 0.030; human activity ≈ 0
G13 ~ mixed — content 0.036*, named 0.066; human activity ≈ 0
G30 ~ mixed — nudge -1.64 min (n 11); content 0.036*; low power
G31 ~ mixed — nudge 0.68 min (n 18); low power
G33 ~ mixed — nudge 0.14 min (n 20); low power
G35 ~ failed — nudge -0.75 min (n 6); low power
G37 ~ mixed — nudge 0.49 min (n 21); low power
G38 ~ mixed — nudge -0.08 [−0.98, 0.82] min (n 82); content 0.020
G39 ~ failed — nudge -1.87 min (n 7); low power
G40 ~ failed — nudge -1.15 min (n 9); low power
G41 ~ mixed — nudge 0.86 [0.51, 1.26] min (n 53); low power
G42 ~ mixed — nudge 0.57 min (n 18); low power
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).



Second figure. Synthetic validation at G51-like sampling. (a) Per-nudge activity response: the H30 estimator recovers zero and is 15–25% low when the response is real (policy-relative). H04’s isolated design is biased by −1.6 min at zero effect. (b) “Same χ every day” tests: Cochran’s Q over-rejects; without day fixed effects, day-level activity shocks fake heterogeneity (60%).

CAVEATS

Only G51 resolves the nudge response; 3–5-day periods give unstable CIs. The G51 level depends on the specification (0.59 pre-registered, 1.04 with day FE; the latter’s pre-window placebo is 0.31). Human activity in dense regime-I chat is not identified. Content uses one embedding model and measures only message-specific steering; human replies may share topics (common cause). Day FE, cluster permutation, fill-phase controls and the splits are post hoc (A2, disclosed). Automated messages stop after 2026-08-20 (undocumented), so the #51 tail may hold no nudges.

ROUND 2

- **Essence.** a trustworthy answer to “will my next message change what this swarm does?”. Round 1 says it is per situation, not a daily state.
- **H30-R1** Situational χ instead of daily χ .
- **H30-R2** The 2026-08-20 switch-off as a natural experiment.
- **H30-R3** Embedding swap and reply control for χ -con.
- **H30-R4** Per-message content strength.
- **H30-R5** Run `confirm.py` after the card is committed.

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